A PROJECT ON

PREDICTIVE ANALYSIS OF CUSTOMER CHURN AND CUSTOMER RETENTION

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Academic project report prepared for learning and demonstration purposes.

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Note: This report uses a clearly labelled illustrative dataset and demonstration figures. It is not based on confidential customer records or a completed live IBM SPSS Modeler run.

1. INTRODUCTION

Customer churn occurs when a customer stops purchasing from, subscribing to, or otherwise maintaining a relationship with an organisation. In subscription businesses, churn may be recorded when a customer cancels a service or does not renew a contract. A high churn rate can weaken recurring revenue, increase the cost of acquiring replacement customers and make business planning less reliable.

Predictive analytics helps organisations identify patterns that may indicate a higher likelihood of customer departure. Rather than waiting until a customer leaves, a business can study service usage, billing behaviour, complaints, contract type and customer engagement to identify accounts that may need attention.

This project presents a learning-oriented case study of **BrightConnect Services**, a fictional subscription-based internet and digital services provider. The proposed analysis uses customer-level attributes to demonstrate how IBM SPSS Modeler can support data preparation, exploration, classification and customer-risk reporting.

The project does not claim that the illustrative values are actual company results. The numbers and charts are included to explain how an analyst could organise and interpret a churn study. A real business should replace the sample data with properly authorised and quality-checked customer records.

Business relevance

A churn-risk score can help a retention team prioritise service calls, investigate recurring network issues, explain billing options and design suitable loyalty offers. The score should support human judgement, not automatically determine customer treatment.

2. OBJECTIVES OF THE STUDY

The main objective is to demonstrate a structured predictive-analytics process for understanding and anticipating customer churn.

Specific objectives

1. Explain customer churn and its effect on recurring business revenue.
2. Identify customer attributes that may be associated with cancellation.
3. Prepare a structured dataset suitable for analysis in IBM SPSS Modeler.
4. Explore churn patterns across contract, service and customer-support groups.
5. Design a classification workflow that estimates the probability of churn.
6. Evaluate the model using measures suited to an imbalanced classification problem.
7. Create practical customer-risk groups for retention planning.
8. Recommend responsible actions that can improve customer experience and retention.

Scope of the study

The scope is limited to a hypothetical subscription-service provider and a demonstration dataset. The analysis focuses on customer status, tenure, contract duration, monthly charges, payment method, service complaints and support contacts. It does not estimate the financial value of individual customers or make claims about any real organisation.

Research questions

- Which customer characteristics should be examined when studying churn?
- How can a classification model support early identification of customers at risk?
- Which model evaluation measures are useful before a retention team relies on predictions?
- How can predicted risk be converted into fair and useful service actions?

3. CONCEPTUAL BACKGROUND AND LITERATURE REVIEW

3.1 Meaning and measurement of churn

Churn is commonly measured as the number of customers lost during a period divided by the number of customers at the start of that period. The exact definition must be fixed before analysis. For example, a business should decide whether temporary suspensions, non-payment closures and customer-initiated cancellations are treated as churn.

3.2 Customer retention

Customer retention refers to the organisation's ability to maintain active customer relationships over time. Retention initiatives may include reliable service, responsive complaint handling, transparent billing, suitable plans and relevant communication. Retention is not achieved by discounts alone; service quality and trust are also important.

3.3 Predictive classification

A classification model learns patterns from labelled historical examples. In a churn project, the target field may contain two classes: Churn and No Churn. Logistic regression, decision trees and other classification algorithms can be considered. The output may include a predicted class and a probability or confidence score.

3.4 IBM SPSS Modeler

IBM SPSS Modeler provides a visual, node-based environment for data preparation and predictive modelling. A typical stream connects data sources, field settings, record operations, modelling nodes and output nodes. This visual structure helps learners document the sequence of analytical steps.

3.5 Why a baseline matters

A model should be compared with a simple baseline, such as predicting the most common class. Without a baseline and a separate validation sample, a high accuracy figure may be misleading, especially when most customers do not churn.

4. RESEARCH METHODOLOGY

The project follows a quantitative, descriptive and predictive approach. It is designed as a reproducible classroom workflow rather than a report of a live company deployment.

Stage	Activity	Expected output
1. Define problem	Specify churn definition and prediction period	Clear target variable
2. Prepare data	Check fields, formats, missing values and duplicates	Analysis-ready file
3. Explore	Review distributions and compare customer groups	Charts and summary tables
4. Train model	Fit a classification algorithm on training data	Model nugget
5. Validate	Test predictions on held-out data	Confusion matrix and metrics
6. Segment	Group customers by estimated risk	Retention priority list
7. Recommend	Translate patterns into service actions	Practical action plan

Software and tools

Microsoft Excel may be used to inspect and organise the sample data. IBM SPSS Modeler is the intended platform for the visual workflow, data auditing, field-role assignment, modelling and output review. Spreadsheet software may also be used to create charts or validate totals.

Sampling and data split

For an actual study, records should be selected from a defined observation window and split into training and testing samples. Where the data are time-dependent, a chronological split is preferable to a random split so that future records are not used to predict the past.

5. DATASET DESCRIPTION AND PREPARATION

5.1 Illustrative dataset

The demonstration data represent a hypothetical set of 900 subscription customers. Each row represents one customer at a defined observation date. The target field, Churn_Flag, records whether the customer cancelled within the chosen prediction window. The following field list is a suggested structure, not a claim that a real customer database was accessed.

Field	Description	Measurement / role
Customer_ID	Unique customer reference	Identifier; exclude as predictor
Tenure_Months	Length of customer relationship	Continuous input
Contract_Type	Monthly, one-year or two-year	Categorical input
Monthly_Charge	Recurring monthly bill	Continuous input
Support_Tickets	Support contacts in observation window	Count input
Late_Payments	Number of late payments	Count input
Service_Issues	Recorded service interruptions	Count input
Auto_Pay	Whether automatic payment is enabled	Categorical input
Churn_Flag	Cancellation during target window	Target: Yes / No

5.2 Data-quality checks

Check duplicate customer IDs, invalid tenure values, inconsistent contract labels, impossible charge amounts and missing target labels. Missing values should be investigated before imputation. Customer identifiers should not be treated as meaningful predictors because they generally encode identity rather than behaviour.

5.3 Data leakage prevention

Only information available at the prediction date should be used as an input. Cancellation reason, final closure date or a post-cancellation survey response must not be included when predicting a future cancellation, because these fields reveal the outcome.

6. PREPARING THE SPSS MODELER STREAM

A recommended visual stream is shown below. The exact node names and available options may vary by installed SPSS Modeler version.

Sequence	Node / operation	Purpose
1	Data source	Read the prepared Excel or CSV file
2	Type	Set measurement levels and target/input roles
3	Data Audit	Inspect distributions, missing values and unusual values
4	Select / Filter	Remove ineligible records and irrelevant fields
5	Partition	Create training and testing samples
6	C5.0 or Logistic Regression	Train a classification model
7	Analysis / Evaluation	Review predictions and classification performance
8	Table / Export	Save scored customer records and summary output

Suggested stream: Source → Type → Data Audit → Select → Partition → Model → Evaluation → Table / Export

Important field settings

Set Churn_Flag as the target and define its values consistently. Contract_Type and Auto_Pay should be treated as categorical fields. Tenure_Months, Monthly_Charge, Support_Tickets, Late_Payments and Service_Issues can be treated as numeric inputs after validation. Customer_ID should be assigned an identifier role or excluded from modelling.

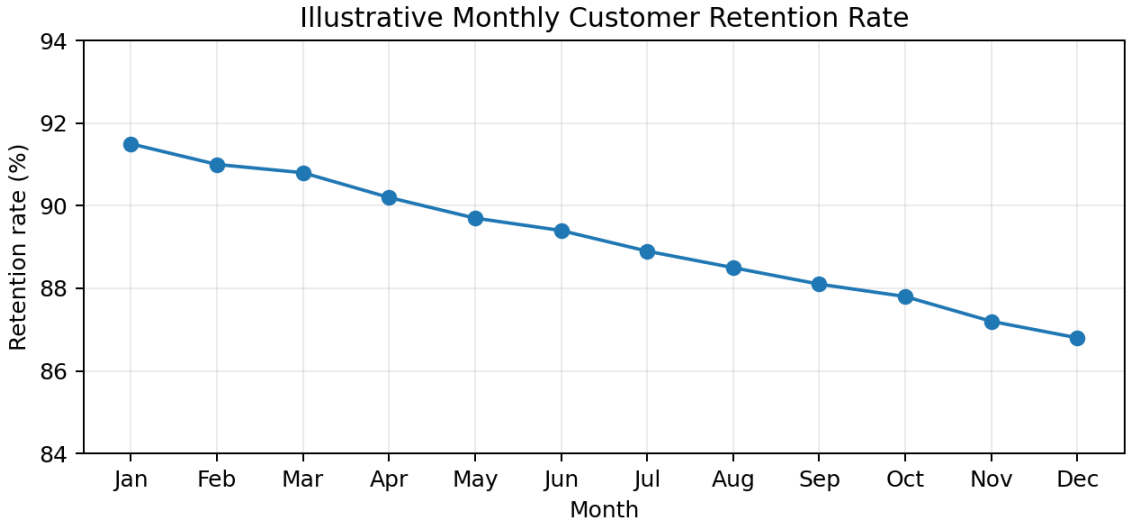
Model execution checklist

Before running the stream, confirm that the target contains both classes, that the partition has no accidental overlap, and that the test data have not been used to tune the model. Save the stream with a clear name and record the software version and modelling settings.

7. EXPLORATORY DATA ANALYSIS

Exploratory analysis helps the analyst understand the customer base before fitting a model. The figures on this page are illustrative teaching examples and are not computed from a live customer file.

7.1 Illustrative retention trend



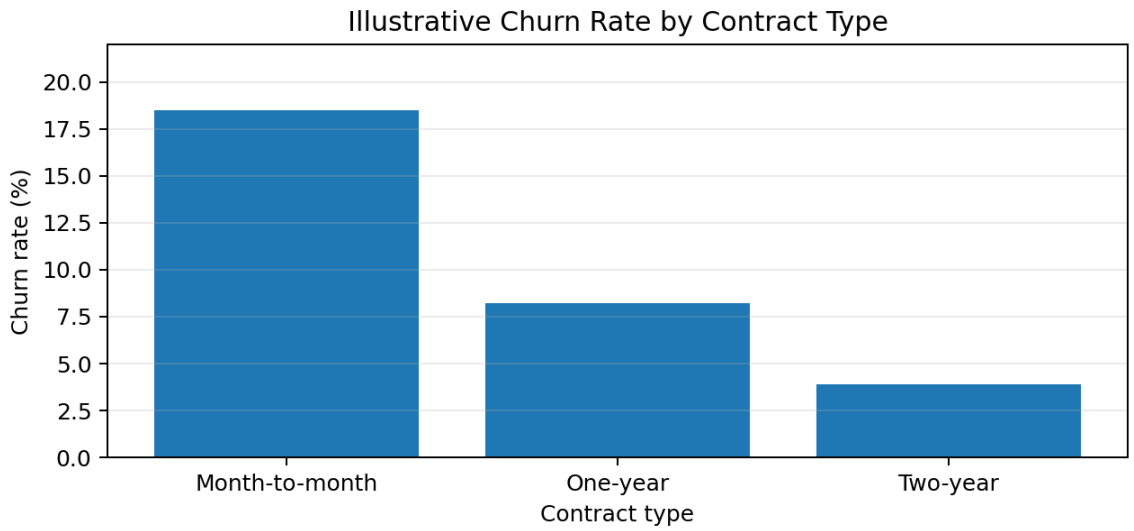
The sample chart depicts a gradual reduction in monthly retention across the year. In a real project, the analyst should confirm the cohort definition, compare equivalent periods and investigate whether the change is linked to service quality, pricing or customer mix.

7.2 Descriptive checks

Useful summaries include customer count, average tenure, median monthly charge, distribution of support tickets, late-payment frequency and churn proportion. Counts and percentages should be shown together so that a small segment is not mistaken for a major source of churn.

8. CUSTOMER GROUP COMPARISONS

8.1 Contract type



The illustrative chart shows higher churn for month-to-month contracts than for longer commitments. This is a pattern to investigate, not proof that contract duration causes churn. Customer needs, pricing, service experience and the reason for choosing a contract may differ across groups.

8.2 Other useful comparisons

Variable	Questions for analysis
Tenure	Are newer customers leaving more often than established customers?
Support tickets	Does repeated contact indicate unresolved problems?
Late payments	Are billing difficulty and cancellation associated?
Service issues	Do repeated outages correspond with higher churn?
Auto-pay status	Does payment convenience differ across churn groups?

The analyst should inspect group sizes and uncertainty before drawing conclusions. Small groups can produce unstable percentages.

9. PREDICTIVE MODEL DEVELOPMENT

9.1 Candidate models

Two suitable starting points are logistic regression and a decision-tree classifier. Logistic regression provides a probability-based model and can be useful for understanding the direction of associations. A decision tree can express rules in an accessible form, such as combinations of short tenure and repeated support contacts. Model choice should be guided by validation performance, interpretability and operational needs.

9.2 Logistic regression concept

For a binary churn outcome, logistic regression estimates the probability of churn from a combination of customer characteristics. The probability is bounded between zero and one. Coefficients describe changes in log-odds, so interpretation should be careful and should consider the scale and coding of each input.

9.3 Decision-tree concept

A decision tree repeatedly splits records into groups that are more homogeneous with respect to the target. It can capture interactions and non-linear patterns, but a deep tree may overfit. Pruning or complexity controls should be considered, and the final tree should be checked on unseen records.

9.4 Training procedure

Fit the candidate model on the training partition only. Review field importance or model coefficients, check for unexpected predictors, and compare alternatives using the same validation approach. If class imbalance is substantial, consider class weights or threshold tuning rather than relying on accuracy alone.

No live model has been executed for this academic report; numerical performance results are therefore not asserted.

10. MODEL EVALUATION AND VALIDATION

A churn model should be evaluated on records that were not used to fit it. A confusion matrix compares predicted and actual outcomes.

	Actual churn	Actual no churn
Predicted churn	True positive (TP)	False positive (FP)
Predicted no churn	False negative (FN)	True negative (TN)

Key measures

Measure	Meaning
Accuracy	Share of all predictions that are correct
Precision	Of customers flagged as churners, share who actually churn
Recall / sensitivity	Of all actual churners, share identified by the model
Specificity	Of customers who do not churn, share correctly identified
F1-score	Harmonic mean of precision and recall
AUC	Ability to rank churners above non-churners across thresholds

Threshold selection

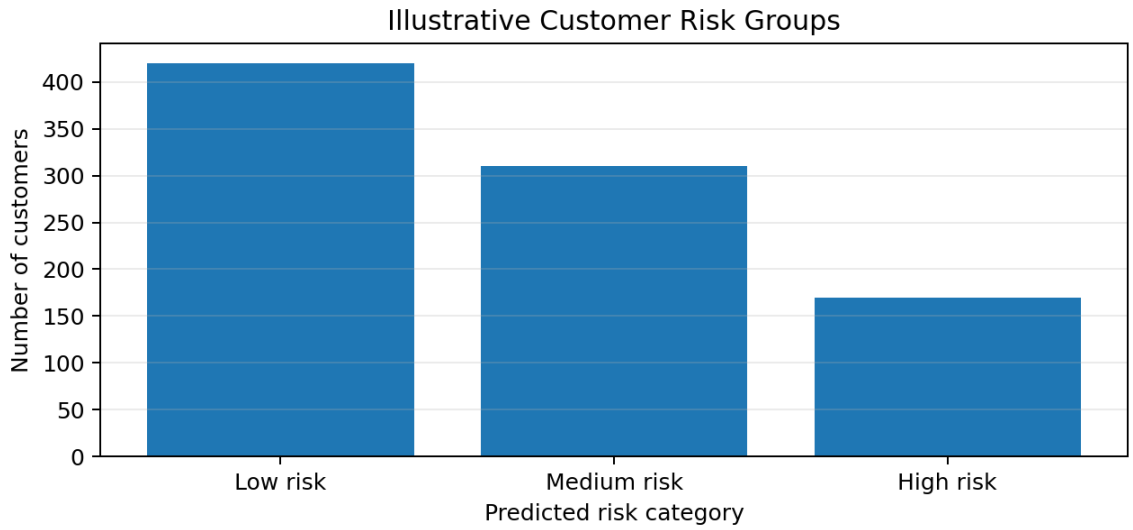
A low threshold may identify more potential churners but also create more false alarms. A high threshold may reduce unnecessary contact but miss customers who need help. The business should select a threshold by considering the cost of missed churn, the cost of retention activity and the customer experience.

Validation safeguards

Check class balance, compare with a baseline, examine performance by important customer groups and monitor results after deployment. A model that performs well overall may still perform poorly for a smaller segment.

11. CUSTOMER CHURN RISK SEGMENTATION

A predicted probability can be converted into broad operational bands. The cut-offs below are examples only and should be selected using validation evidence and the capacity of the retention team.



Risk band (example)	Illustrative score	Possible response
Low	Below 0.30	Continue normal service and routine communication
Medium	0.30 to 0.60	Review service history and offer self-service support
High	Above 0.60	Prioritise a helpful human review and resolve open issues

12. RECOMMENDATIONS

1. Improve onboarding: contact new customers early and explain setup, billing and support channels.
2. Resolve recurring faults: identify repeated service issues and assign clear ownership for resolution.
3. Make billing transparent: provide clear invoices, payment reminders and accessible payment options.
4. Review contract experience: explain plan choices without pressuring customers into longer commitments.
5. Use targeted retention contact: prioritise customers based on validated risk and relevant service context.
6. Track outcomes: measure whether interventions improve retention, not just how many calls are made.
7. Refresh the model: retrain and validate periodically as customer behaviour and products change.

13. LIMITATIONS AND ETHICAL CONSIDERATIONS

The dataset and charts in this report are illustrative. They cannot establish actual churn rates, causal relationships or model accuracy for a real service provider. A real study would require authorised data, a documented churn definition, sufficient observations and validation on an appropriate hold-out period.

Customer data may contain personal or commercially sensitive information. Access should be limited to authorised staff, identifiers should be removed or protected where possible, and retention actions should be fair and transparent. Predictions should not be used to deny essential support or impose unfair prices. Human review and a clear route for customers to correct inaccurate information are important safeguards.

14. CONCLUSION

This project outlines how predictive analytics can support customer retention planning in a subscription-based business. It explains the meaning of churn, proposes a suitable dataset, describes a node-based IBM SPSS Modeler workflow and introduces exploratory analysis, classification and model evaluation. It also shows how a validated probability score may help a retention team prioritise customer support.

The central lesson is that prediction is only one part of retention management. Reliable data, a clearly defined outcome, careful validation, responsible use and genuine improvements in customer experience are all necessary. The proposed workflow can be adapted to real organisational data after appropriate approval and testing.

REFERENCES AND LEARNING RESOURCES

• IBM. IBM SPSS Modeler documentation and product learning resources.
• Han, J., Kamber, M. and Pei, J. *Data Mining: Concepts and Techniques*. Morgan Kaufmann.
• James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*. Springer.
• Provost, F. and Fawcett, T. *Data Science for Business*. O'Reilly Media.
• Course materials on predictive analytics and customer relationship management.

Declaration: The student should review, verify and adapt this illustrative academic draft to the university's submission requirements and should not present simulated data or unexecuted model results as actual research findings.